

PART A [25 marks]

This part consists of **3** questions. Answer **all** questions.

1. (a) Find $\lim_{x \rightarrow \infty} \frac{5x^3 + 4x - 9}{7 - 3x^3}$. [4 marks]
- (b) The function $g(x)$ is defined by

$$g(x) = \begin{cases} m^2 - 8 & , x \leq 2, \\ \frac{x-2}{\sqrt{2x}-2} & , 2 < x \leq 8 \end{cases}$$
 where m is a constant. If $\lim_{x \rightarrow 2} g(x)$ exists, find the values of m . [5 marks]
2. (a) Find the derivative of $f(x) = \frac{1}{x+3}$ by using the first principle. [6 marks]
- (b) Given $y = x^2 e^{-3x}$, find $\frac{d^2 y}{dx^2}$. Give your answer in the simplest form. [5 marks]
3. Given $f(x) = x^2 - \frac{6}{x}$, where $x \neq 0$. Find the coordinate(s) of stationary point and determine whether it is a relative maximum or a relative minimum. Give the coordinates correct to 3 decimal places. [5 marks]

PART B [75 marks]

This part consists of **7** questions. Answer **all** questions.

1. Given the complex numbers $z_1 = -i$ and $z_2 = 2 + i\sqrt{3}$.
 - (a) Express z_1^2 and $\bar{z}_2$ in the form $a + bi$ where $a, b \in R$ [3 marks]
 - (b) From part 1(a), find $W = \frac{z_1^2 + \bar{z}_2}{z_1}$. Hence, find $|W|$ and $\arg W$. [7 marks]
2. Solve the following:
 - (a) $\log_2 2x = 2\log_4 (x+4)$. [5 marks]
 - (b) $2 \left| \frac{x-3}{2x-1} \right| \geq 1$. [6 marks]
3. Given $f(x) = \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}$

- (a) Simplify $f(x)$ and hence, evaluate $f\left(\frac{1}{2}\right)$. [4 marks]
- (b) The domain of $f(x)$ is a set of real number except three numbers. Determine the numbers. [4 marks]
4. Given a function $f(x) = \ln(2x + 1)$.
- (a) State the domain and range of $f(x)$. [2 marks]
- (b) Find the inverse function of $f(x)$ and state its domain and range. Hence, find the value of x for which $f^{-1}(x) = 0$. [5 marks]
- (c) Sketch the graphs of $f(x)$ and $f^{-1}(x)$ on the same coordinates axes. [3 marks]
5. (a) The function $g(x)$ is defined by
- $$g(x) = \begin{cases} 3 & , 2 < x \leq 8 \\ \frac{|8-x|}{x-8} + k & , x > 8. \end{cases}$$
- Find the values of k such that $g(x)$ is discontinuous at $x = 8$. [4 marks]
- (b) A function $f(x)$ is defined as
- $$f(x) = \begin{cases} \frac{9-x^2}{x-3} & , x < 3, \\ -6 & , x = 3, \\ \frac{162x-54x^2}{x^3-27} & , x > 3. \end{cases}$$
- Determine whether $f(x)$ is continuous at $x = 3$. [7 marks]
- 6 (a) Given $x^2y = \sin(2x^2 + \pi)$. Show that $2y(1+8x^4) + 4x\frac{dy}{dx} + x^2\frac{d^2y}{dx^2} = 4\cos(2x^2 + \pi)$. [7 marks]
- (b) Consider the parametric equations $x = 2t - t^{-1}$, $y = 2t + t^{-1}$
- (i) Show that $\frac{dy}{dx} = \frac{2t^2-1}{2t^2+1}$. [4 marks]
- (ii) Find $\frac{d^2y}{dx^2}$ in term of t . Hence, show that $\frac{d^2y}{dx^2} = \frac{8}{y^3}$. [5 marks]

7. Given a right circular cylinder with height $2h$ and radius $r = \sqrt{a^2 - h^2}$, which is inscribed in a sphere of fixed radius a .
- (a) Show that the volume of the cylinder is $V = 2\pi(a^2 - h^2)h$. Hence, find the first and second derivative of the volume, V . [5 marks]
- (b) From part 7(a), find the maximum value of the volume, V in terms of π as the height, h varies. [4 marks]

END OF QUESTION PAPER

Final Answer

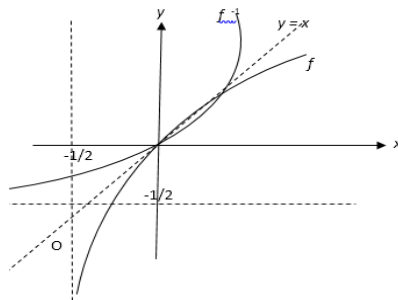
PART A

1. (a) $-\frac{5}{3}$
(b) $\pm\sqrt{10}$
2. (a) $-\frac{1}{(x+3)^2}$
(b) $e^{-3x}(9x^2 - 12x + 2)$
3. $(-1.442, 6.240)$ is the minimum point

PART B

1. (a) $z_1^2 = -1, \bar{z}_2 = 2 - i\sqrt{3}$
(b) $W = \sqrt{3} + i, |W| = 2, \arg(W) = \frac{\pi}{6} @ 0.524$
2. (a) $x = 4$
(b) $\left(-\infty, \frac{1}{2}\right) \cup \left(\frac{1}{2}, \frac{7}{4}\right]$
3. (a) $\frac{x+1}{2x+1}, \frac{3}{4}$
(b) $-1, -\frac{1}{2}, 0$
4. (a) $D_f = \left(-\frac{1}{2}, \infty\right), R_f = (-\infty, \infty)$
(b) $f^{-1}(x) = \frac{e^x - 1}{2}, D_{f^{-1}} = (-\infty, \infty), R_{f^{-1}} = \left(-\frac{1}{2}, \infty\right), x = 0$

(c)



5. (a) $k \neq 2$
(b) $f(x)$ is continuous at $x=3$.
6. (a) Shown
(b) (i) Shown (ii) $\frac{8t^3}{(2t^2+1)^3}$
7. (a) $\frac{dV}{dh} = 2\pi a^2 - 6\pi h^2, \frac{d^2V}{dh^2} = -12\pi h$
(b) $\frac{4\sqrt{3}\pi a^3}{9}$